

MANAV NARULA

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Professional Summary

AI Engineer with 2 years of experience building production-grade Generative AI, Computer Vision, and backend systems. Experienced in developing multimodal LLM pipelines, OCR and document-processing workflows, RAG applications, and AI agents using Python, OpenAI, Gemini, LangChain, and LangGraph. Hands-on experience building FastAPI/Flask services, integrating AI systems with external APIs, and deploying applications using AWS, Docker, and Linux.

Work Experience

Billtech Labs

Sep 2024 - Present

AI Engineer

- Developed a multimodal AI pipeline using **GPT-4o** for multimodal extraction and semantic parsing, extracting **17+ structured attributes** from **300+ jewellery CAD images daily** and generating automated JSON and Excel outputs via **Pandas**.
- Contributed to the integration and testing of a multimodal jewellery visual-search system using **SigLIP 2** and **DINOv2** embeddings, supporting image/text queries and **GPU-accelerated** inference.
- Built a **FastAPI** backend integrated with **Meta WhatsApp Cloud API** webhooks to handle image/document ingestion, format conversion, and asynchronous delivery of processed outputs to end users.
- Engineered OCR pipelines using **EasyOCR** and **OpenCV** to extract text with bounding boxes and confidence scores, followed by **LLM-based** structuring into JSON and automated Excel reports.
- Implemented **GPT-based** query understanding for spelling correction, category classification, and query rewriting, and integrated **Gemini Vision API** for automated jewellery image background removal and enhancement.
- Deployed and maintained services on **AWS EC2** using **Docker** and Linux.

Projects

Autonomous AI Career Agent | Python, LangGraph, OpenAI API, FastAPI, SQLite, Langfuse

[GitHub](#)

- Built an autonomous career agent using a **LangGraph** plan-route-execute-reflect workflow with 6 specialized tools for resume analysis, skill-gap detection, job search, JD matching, learning roadmaps, and interview preparation.
- Shipped a **FastAPI** backend with SQLite persistence, step/token cost guardrails, and **Langfuse** tracing; added PDF report generation and an evaluation suite achieving a **90%** evaluation score on a labeled resume-JD matching dataset.

Hybrid RAG + AI Agent System | Python, LangChain, OpenAI API, FAISS, Pandas

[GitHub](#) | [Live Demo](#)

- Built a hybrid RAG system using **LangChain** and **OpenAI** that routes queries between **FAISS-based** retrieval and a **Pandas DataFrame agent** based on query intent.
- Implemented multi-format ingestion for PDF, TXT, and CSV files with LLM-based document classification for invoices, contracts, and bank statements.

Education

Guru Gobind Singh Indraprastha University

2021 - 2024

Bachelor of Computer Applications (BCA)

New Delhi

Technical Skills

- Generative AI & LLMs:** OpenAI API (GPT-4, GPT-4o), Anthropic Claude API, Google Gemini API, Prompt Engineering, Structured Outputs, Function Calling, RAG, LLM Evaluation
- AI Agents & Retrieval:** LangGraph, LangChain, FAISS, Vector Databases, Embeddings, Tool Calling, Query Routing, Langfuse
- Computer Vision & OCR:** PyTorch, Hugging Face Transformers, OpenAI Vision API, EasyOCR, OpenCV, Image Processing
- Languages:** Python, SQL
- Backend Development:** FastAPI, Flask, REST APIs, Webhooks, Pydantic, Asynchronous Processing, SQLite
- Cloud & Tools:** AWS EC2, Docker, Linux, Git, WhatsApp Cloud API, Pandas, Streamlit
- Machine Learning:** Scikit-learn, Classification, Regression, Feature Engineering, Model Evaluation